

Unit-II

Nutrient

Nutrients that are required in vast amounts are called macronutrients. There are three types of macronutrients:

Carbohydrate, protein, fat

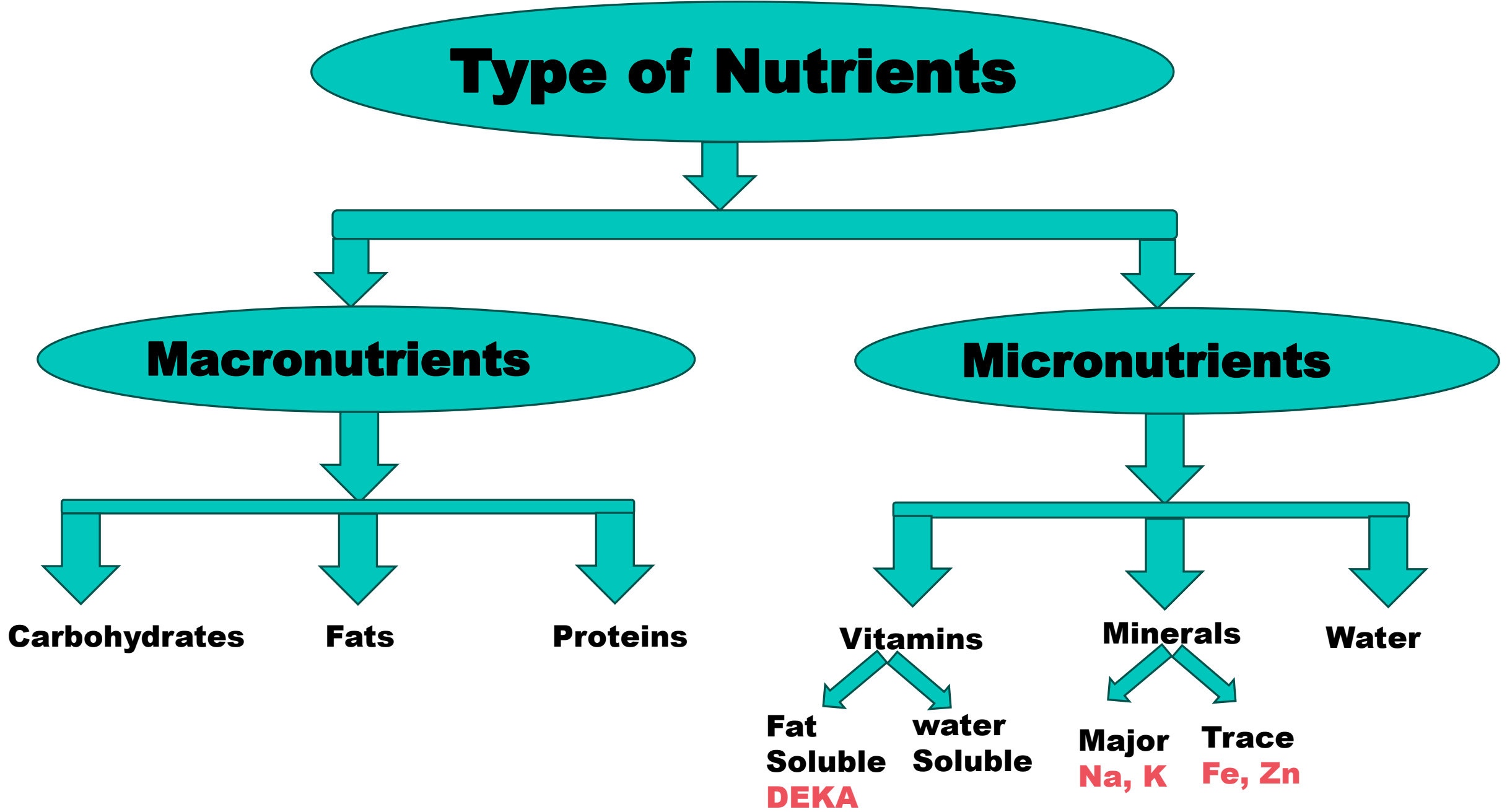
Macronutrients are mostly carbon-based compounds that can be metabolically processed into cellular energy through changes in their chemical bonds. The chemical energy is transformed into cellular energy known as ATP, which is used by the body to carry out basic functions.

The quantity of energy a person consumes on an everyday basis originates basically from the 3 macronutrients carbohydrates, protein, fat. Food energy is measured in calories or kilocalories.

Nutrients that are required in small amount are called micronutrients.

Vitamin, mineral, water

Type of Nutrients



```
graph TD; A([Type of Nutrients]) --> B[Macronutrients]; A --> C[Micronutrients]; B --> D[Carbohydrates]; B --> E[Fats]; B --> F[Proteins]; C --> G[Vitamins]; C --> H[Minerals]; C --> I[Water]; G --> J[Fat Soluble DEKA]; G --> K[water Soluble]; H --> L[Major Na, K]; H --> M[Trace Fe, Zn];
```

The diagram is a hierarchical flowchart. At the top is a teal oval labeled 'Type of Nutrients'. A teal arrow points down from this oval to a horizontal teal bar. From this bar, two teal arrows point down to two teal ovals: 'Macronutrients' on the left and 'Micronutrients' on the right. From the 'Macronutrients' oval, a teal arrow points down to another horizontal teal bar. From this bar, three teal arrows point down to the text labels 'Carbohydrates', 'Fats', and 'Proteins'. From the 'Micronutrients' oval, a teal arrow points down to another horizontal teal bar. From this bar, three teal arrows point down to the text labels 'Vitamins', 'Minerals', and 'Water'. From the 'Vitamins' label, two teal arrows point down to 'Fat Soluble' and 'water Soluble'. Below 'Fat Soluble' is the text 'DEKA' in red. From the 'Minerals' label, two teal arrows point down to 'Major' and 'Trace'. Below 'Major' is the text 'Na, K' in red. Below 'Trace' is the text 'Fe, Zn' in red.

Macronutrients

Carbohydrates

Fats

Proteins

Micronutrients

Vitamins

**Fat
Soluble
DEKA**

**water
Soluble**

Minerals

**Major
Na, K**

**Trace
Fe, Zn**

Water

Protein

Proteins were first described by the Dutch chemist Gerardus Johannes Mulder and named by the Swedish chemist *Jöns Jacob Berzelius* in 1838.

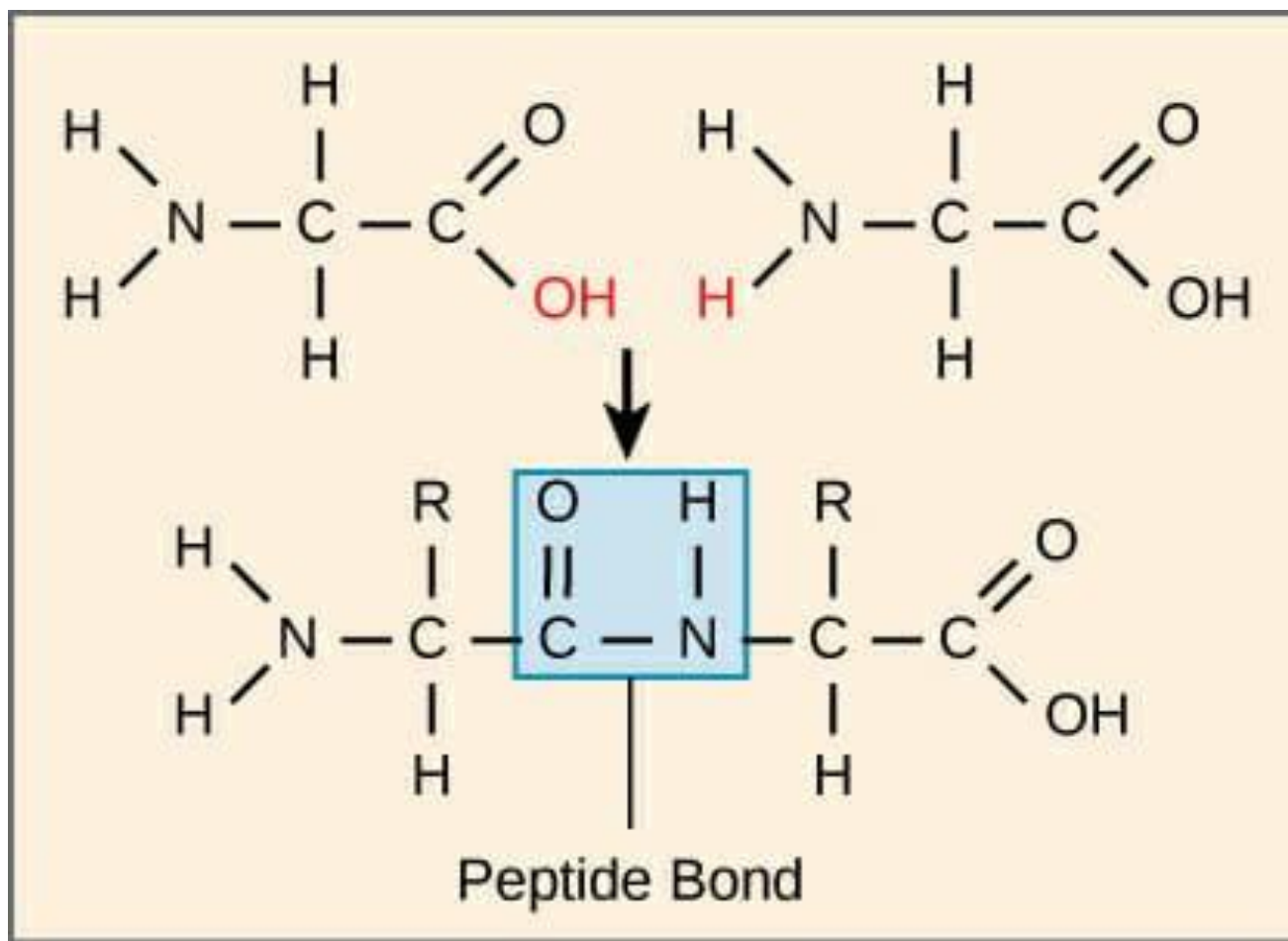
Chemical Organization- Protein is an organic compound. It contains carbon, hydrogen, oxygen, and nitrogen. Apart from these, sulphur and phosphorus are also present in some protein molecules. Iron, copper, cobalt, and other salts are also present along with some protein molecules. Nitrogen is the main component of protein.

Amino Acids—Protein breaks down into its simplest unit amino acids after hydrolysis. Therefore, the amino acid is the smallest unit of protein. Many amino acids combine to form proteins.

Protein molecules are very large. Their molecular weight ranges from 15,000 – 50,000.

Definition-Proteins are the main constituent of all the body cells and thus form the greater part of muscles and other tissues.

A total of 23 amino acids are found in the food substance, and more than 120 amino acids in the protein are joined by a peptide linkage to form a protein molecule.



Classification of proteins

Proteins are large molecules formed by the combination of a number of amino acids that have been found to occur in proteins and are important from the point of view of human nutrition. Amino acids can be classified as follows:

<i>Essential</i>	<i>Semi-essential</i>	<i>Non-essential</i>
Histidine	Arginine	Glutamic acid
Lysine	Tyrosine	Aspartic acid
Tryptophan	Cystine	Alanine
Phenylalanine	Glycine	Proline
Methionine	Serine	Hydroxyproline
Threonine		Cysteine
Leucine		
Isoleucine		
Valine		

Classification of Proteins (Based on chemical composition)

Simple proteins: It is composed entirely of amino acids only.

Conjugated or Complex proteins: It is made up of amino acids and other organic or inorganic compounds.

The non-amino acid group is termed as Prosthetic group (e.g.). Lipoproteins—Chylomicrons.

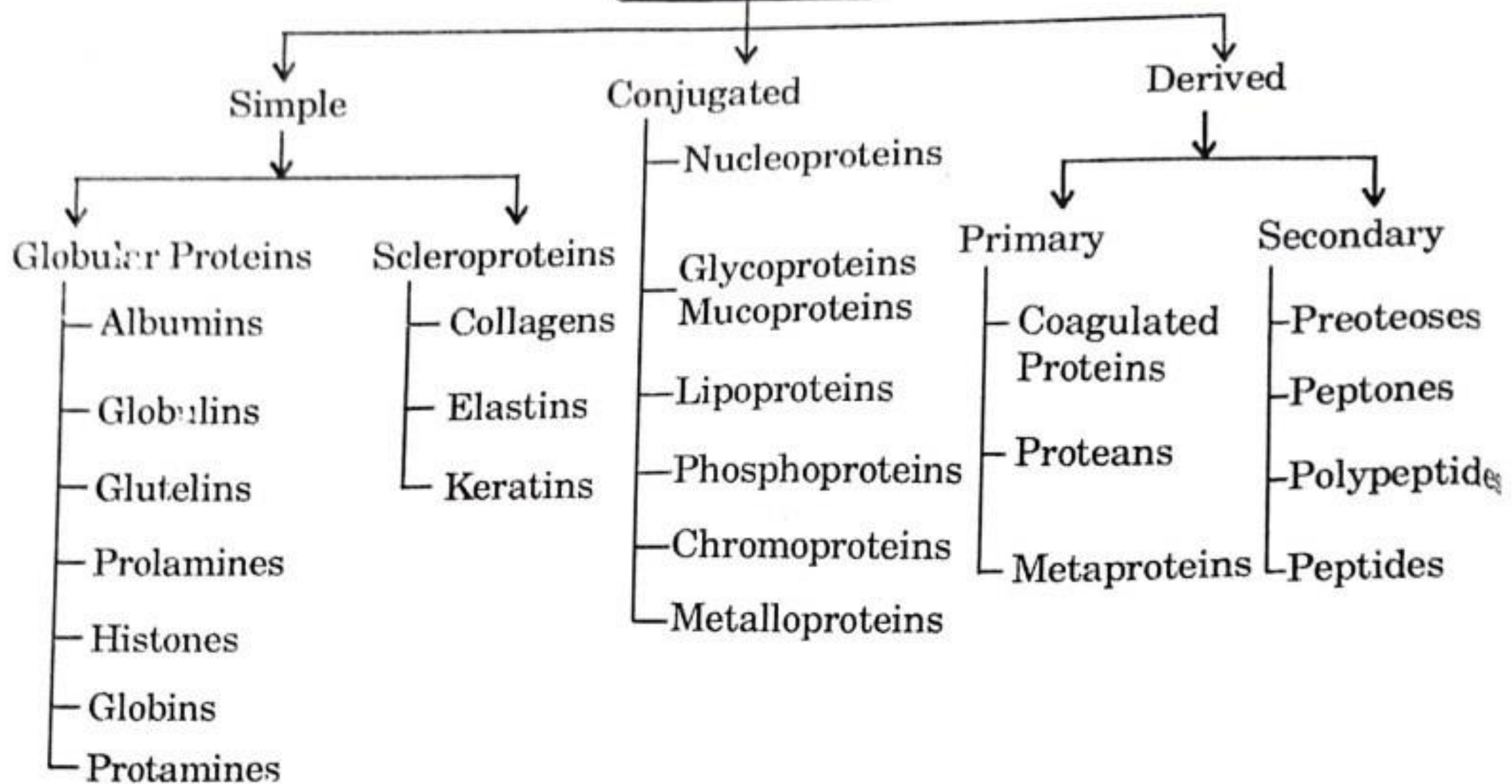
Derived proteins: These are derivatives of proteins resulting from the action of heat, enzymes, or chemical reagents. This group also includes artificially-produced polypeptides e.g. Fibrin.

Classification of proteins (Based on nutritional value)

Proteins are classified into two types based on nutrition viewpoint as follows:

Complete proteins: These contain all the essential amino acids in sufficient quantity to supply the needs of the body. They support life even if supplied as the sole source of protein. These proteins are of animal origin (e.g.) milk, meat, poultry, egg, and fish. The quality of these proteins is much superior to those of incomplete proteins.

PROTEINS



Incomplete proteins: These proteins are deficient in one or more of the essential amino acids and therefore, they do not support life on their own. All plant sources of proteins (i.e.) vegetables, fruits, cereals, pulses, nuts, and oilseeds contain incomplete proteins to varying degrees.

COMPLETE VS. INCOMPLETE PROTEINS

Dietary protein is required for the body as there are 9 essential amino acids the body cannot create and must obtain from one's diet. Complete proteins contain all 9 of these essential amino acids versus incomplete proteins do not.

Complementary proteins are combinations of two or more incomplete proteins that supply all 9 essential amino acids.

Complete Proteins:

- Meat
- Poultry
- Dairy
- Eggs
- Fish

Incomplete Proteins

- Vegetables
- Grains
- Legumes/Beans
- Nuts/Seeds

Complementary proteins: If two sources of incomplete proteins are combined in the same meal, the resulting protein may be of better quality. These are called complementary proteins e.g. Pongal prepared using moong dhal and rice is of better quality than rice or dhal cooked separately. Rice is deficient in the amino acid lysine but rich in methionine. Pulses are rich in lysine but deficient in methionine. So, rice and pulse combination will complement each other. Rice Kheer is another example, where animal and vegetable proteins -milk and rice are cooked together.

Complementary Proteins:

→ Grains+Pulses/Vegetables

→ Nuts/Seeds+Vegetables/Pulses

Food sources of proteins

Milk is a valuable source of protein (**casein**) although it does not contain a large quantity of protein, the quality is excellent.

Good sources of plant proteins are pulses, nuts, and oilseeds.

CLASSIFICATION OF PROTEIN ACCORDING TO MERITS OF PHYSICAL AND CHEMICAL PROPERTIES

Proteins are classified into three major categories according to their physical and chemical properties, namely

(1) Simple Proteins-Simple proteins are those proteins that are converted into amino acids only after the body's decomposition (Hydrolysis). The following proteins are simple proteins such as:

(i) Albumin: It dissolves in water and coagulates by heat. The lacto albumin of milk is an example,

(ii) Globulin: It is soluble in alkali solution and gets solidified by heat, Such as carcinogens of milk, ovoglobulin of eggs, tuberin of potatoes, etc.

(iii) Glutelin: It is soluble in mild acids and bases. For example, the gluten of wheat, arikshinin of

(iv) Prolamin: It is soluble in only 70-80% alcohol. Example: Blydin of wheat, Zein of maize.

(v) Fibrous Protein: These are **not soluble in any solution**, they are found in the structure of bones, skin, nails, and hair. For example, collagen of tissue, keratin of hair, and myosin of muscle.

(2) Conjugated Protein- Simple protein when combined with another non-protein component, then it is called conjugated protein. Examples of this are the following:

(i) Nucleoprotein—It is found in every cell.

(ii) Glycoprotein-Glycoprotein is found in globulin and albumin of blood.

(iii) Phosphoprotein- is found in milk carcinogen and egg yolk in the form of ovoprotein.

(iv) Haemoglobin-found in the blood by the name of hemoglobin.

(3) Derived Protein-Due to the breakdown of simple and combined proteins in the digestive process, two types of derived proteins are formed before the formation of amino acids. They are primary and secondary derived proteins.

FUNCTIONS OF PROTEIN

Proteins perform the following functions in our body.

- 1. The function of body building**-Protein is mainly a **body-building food element**. The presence of protein is followed by the amount of water in the body. For example, protein is mainly present in hair, nails, skin, endocrine glands, muscles, and bones, etc.

Protein is found in all the fluids of the body except bile and urine. It plays an important role in **forming new tissues**, whereas in infancy, childhood, adolescence, and pregnancy, protein is the main requirement for physical growth and development.

- 2. Work for the reconstruction of the body's wear and tear**-Just as protein helps in the formation of new tissues, in the same way, protein is needed for the repair of the wear and tear caused by the continuous functioning of the body. The wear and tear of cells and tissues in the body can occur during an accident, injury, burn or operation, etc.

3. Blood plasma for a clotting- A protein called fibrinogen is found in the blood plasma for clotting of blood. Bleeding occurs due to injury, cut, or accident. If the blood does not clot within 4-5 minutes and the bleeding does not stop, a lot of blood is lost from the body and life can be in danger. Blood clotting is a complex process. In this process, thrombin, which is a protein converts fibrinogen to fibrin. Due to this, the flow of blood stops. (fibrinogen)

4. Regulating the various functions of the body- Protein keeps balanced between the amount of acid and base in our body regular and when needed, the protein itself acts like a base or acid, which is called Protein Buffers. It does not allow the acid base of the body to accumulate so that the body's pH can remain stable.

5. Regulating the balance of water in the body- Due to the lack of protein, water starts accumulating in the body, because protein maintains the balance of water content on both sides by controlling the activities inside and outside the cell.

6. Producing enzymes, hormones, antibodies- Proteins form many enzymes in the body. It forms many nitrogen-containing compounds in the body, by which many chemical reactions are completed.

(a) Manufacture of enzymes—These are made of proteins and proteins form enzymes. Enzymes perform various functions in the body; For example, digestion of food, oxidation, metabolism, etc. Enzymes like pepsin, amylase, lipase, etc. are produced only in the presence of proteins. Proteins form many nitrogen-rich compounds in the body.

(b) Producing Hormones-Proteins are essential for the manufacture of hormones. Hormones control and regulate various activities. For example, the **thyroxine hormone** regulates the oxidation process, physical growth, **insulin** plays a main role in maintaining the level of glucose in the blood. Similarly, the pituitary, adrenaline, gender hormones, etc. perform various functions in the body.

(c) In the formation of antibodies- Antibodies provide the body's ability to fight against various diseases and help in increasing the immunity of the body. **Antibodies are not produced due to a lack of protein in the diet.** As a result, the ability of a person to fight diseases decreases. A person suffers from many types of diseases and falls ill again and again. Therefore, it is very important to have protein in the food to lead a healthy life.

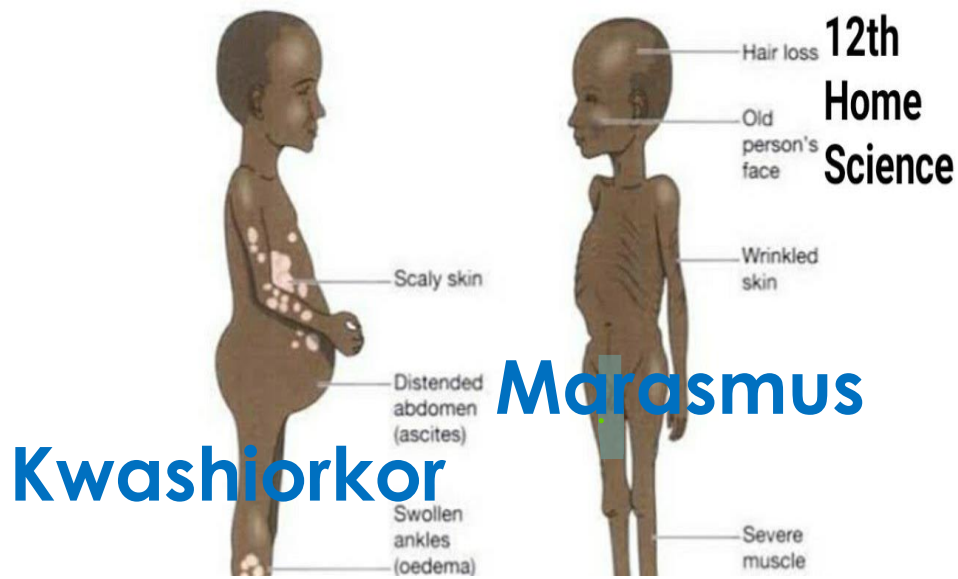
- 8. In providing energy-** in the absence of carbs and fats, protein also serves to provide energy. One gram of protein provides 4 calories of energy. Whenever there is not enough fat and carbs present in the diet, then only protein provides energy. When protein is used in excess, it accumulates in the body in the form of fat and when these are used to provide energy, body-building, which is the main work, becomes secondary.
- 9. In milk production-** The mother's milk is the first food of a newborn baby. Mother's milk is the best for the proper nutrition of the baby. Protein is essential for the production of breast milk. Since mother's milk **contains 1.2% protein**. Therefore, the lactating mother should consume more amount of protein.

EFFECT OF PROTEIN DEFICIENCY

Protein is very essential for physical growth and development. Protein is the body's constructive element and makes new cells, **fibers, and tissues and repairs damaged cells and fibers that are broken.** A deficiency of protein leads to **anemia, marasmus, kwashiorkor, diarrhea, bloating, etc.** The liver does not perform its function properly in the absence of protein.

Most of the victims of protein deficiency are children of 1-5 years. Protein-calorie malnutrition occurs due to a lack of both protein and energy. The following three diseases mainly occur in children due to a deficiency of protein.

- 1) **Kwashiorkor**-Kwashiorkor was discovered by Sisley William in 1935. It means “the disease that occurs to the first child after the birth of the second child.” The following symptoms are seen in children suffering from Kwashiorkor:
- 2) **Marasmus**—This disease occurs due to a deficiency of both protein and energy. After the age of 6 months, apart from the mother's milk, other food items are also necessary for the children, but if the children do not get them, there is a deficiency in the nutritional elements.



KWASHIORKOR

- ✓ Acute illness/infections, measles, AGE, trauma, sepsis are some causes
- ✓ Protein is principal nutrient
- ✓ 18 months to 3 years
- ✓ Rapid, acute onset
- ✓ Some weight loss
- ✓ High mortality

MARASMUS

- ✓ Severe prolonged starvation, chronic/recurring infections
- ✓ Calories and protein are principal nutrients
- ✓ 6 months to 2 years
- ✓ Chronic, slow onset
- ✓ Severe weight loss
- ✓ Low mortality unless related to underlying disease condition

(3) Wasting of Muscles—The muscles of the body start getting destroyed. Its effect is more on the muscles of the arms, hands, and legs. The arms, hands, and legs become thin and weak.

(4) Mental changes-Apathy and irritability come in the nature of the child. Children start crying while talking and they do not want to play the child feels sluggish, lethargic, and tired.

(5) Anemia-Due to lack of protein, hemoglobin is not formed. Hence. there is a disease of blood deficiency.

(6) Effect of fat in the liver-Due to the lack of protein, fat deposits on the liver, due to which its size increases. The size of the pancreas becomes smaller.

7) Effects on the skin-Due to lack of protein, the skin becomes dry and dull. There is a change in the color of the skin. There are dark brown and black marks on the skin.

(8) Effect on hair-Hair becomes dry, rough, and dull. There is a change in the color of the hair. Hair turns gray or white, and hair growth stops.

(9) Loss of Appetite and Diarrhoea-Due to lack of protein, many digestive disturbances arise. Many digestive enzymes are not produced. The child doesn't feel hungry and diarrhea may also occur.

(10) Effects on Mucous Membrane-Mucous membranes are affected due to lack of protein. Due to this, the lips begin to crack and the surface of the tongue becomes greasy.

(11) Moon-shaped Face—In addition to the hands and feet being swollen, the swelling also occurs on the face, due to which the child's face becomes round like the moon.

(12) Effect on Nervous System-Due to lack of protein, not only the physical development of the child stops, but mental development also stops. The child's ability to learn decreases. Endocrine hormones are not produced and secreted properly. There is a disturbance in blood circulation.

(13) Vitamin Deficiency—Protein has a role in the formation of some vitamins. Due to the deficiency of B vitamins, there is a burning sensation in the corners of the lips and in the tongue, due to which the whole mouth is not able to open.

(14) Growth stops—Some children have an increased appetite, and some have a low appetite. Due to an increase or decrease in appetite, the growth of the body stops, and the stomach contracts. The weight and height of the child decrease. Dwarfism occurs in the child.

(15) Digestive system is affected—In by this disease, loose watery diarrhea occurs and the stomach becomes bloated.

(16) Inflammation in the body—Swelling occurs in the body due to lack of protein. First, there is swelling in the legs, then there is swelling in the body.

Excess of Protein

1.Kidney Strain: Excessive protein intake can put a strain on the kidneys as they are responsible for filtering and eliminating waste products from protein metabolism, including nitrogen. Over time, this increased workload can lead to kidney damage or dysfunction.

2.Dehydration: High-protein diets often result in increased water loss through urine. This can lead to dehydration if adequate fluid intake is not maintained.

3.Mineral Imbalances: High protein intake can disrupt the balance of minerals in the body, particularly calcium. It may lead to increased calcium excretion through urine, potentially contributing to a **risk of osteoporosis and kidney stones**.

4.Digestive Issues: Excess protein can be hard on the digestive system, potentially causing constipation, bloating, and other gastrointestinal discomfort.

5. Increased Risk of Heart Disease: Some high-protein diets, particularly those high in animal protein, may also be high in saturated fats and cholesterol, increasing the risk of heart disease.

6. Weight Gain: Contrary to some beliefs, excessive protein consumption alone may not necessarily lead to weight loss. If the excess protein intake is not balanced with other nutrients and calorie control, it can contribute to weight gain.

7. Bad Breath and Body Odor: Excess protein metabolism can produce ammonia, which can lead to bad breath and unpleasant body odor.

8. Liver Disorders: In rare cases, excessive protein intake can stress the liver and potentially lead to liver disorders.

9. Nutrient Imbalances: Overemphasis on protein intake can sometimes lead to a lack of other essential nutrients, as individuals may prioritize protein-rich foods over a balanced diet containing fruits, vegetables, and whole grains.

DAILY REQUIREMENT OF PROTEIN

The demand for protein is high in infants, children, adolescents, and pregnant and lactating mothers. The demand for protein also increases in the elderly. Indian Council of Medical Research (ICMR) in 1933) nutritionists proposed the daily requirement of protein for Indians as follows:

Daily Requirement of Protein

<i>Stage</i>	<i>Protein (grams per day)</i>
1. Infancy	
0-6 months	2.05 g/kg by weight
6-12 months	1.65/kg by weight.
2. Childhood	
1-3 Years	22
4-6 Years	30
7-9 Years	41
10-12 Years	54
3. Adolescence	
13-15 Years (Girls)	65
13 - 15 Years (Boys)	70
16 - 18 Years (Girls)	63
16-18 years (boys)	78
4. Adult male	60
5. Adult female	
Low active	
Moderate active	50
More active	
6. Pregnancy	50 + 15 g
7. Lactation stage	
0-6 months	50 + 25 = 75 g
6-12	50 + 18 = 68 grams

Protein content in various food items

<i>Food Ingredients (per 100 g)</i>	<i>Protein content (in gram)</i>	<i>Food Ingredients (per 100 g)</i>	<i>Protein content (in gram)</i>
Animal food		Dried Nuts	
Fish	21.8	Groundnut	31.5
Meat	19.8	Cashews	21.2
Liver	19.3	Almonds	18.3
Eggs	13.3	Walnuts	15.6
Milk and Milk Made		Dry Coconut	6.8
Buffalo's Milk	4.3	Fresh Coconut	4.2
Cow's Milk	3.2	Moong Dal	24.5
Mother's Milk	1.1	Arhar Dal	24.0
Curd	3.2	Chickpeas	20.0
Mawa	20.1	Soyabean	43.2
Paneer	25.0	Cowpea	24.3
Lentil		Rajma	23.0
Lentils	25.1	Black Gram	24
Wheat		Dry Pea Cereal	19.7
Wheat Flour	12.0	Green Peas	2.0
Maize	11.0	Sweet Potato	1.9
Semolina	10.6	Beet	1.8
Maida	10.1	Potato	1.6
Rice	7.6	Onion	1.2
Vegetables		Carrot	1.00
Turnip Leaves	2.9	Tomato	1.00
Radish Parima	2.7		
Banana	1.00		
Radish	0.7		
Gourd	0.4		

FACTORS AFFECTING THE PROTEIN'S DEMAND

Body –The greater the breakdown in the cells, and tissues of the body the more the muscle is damaged. More protein is needed to repair and rebuild these broken damaged cells and tissues.

Gender-The demand for protein is more in women than in men. Every month women menstruate, due to which a lot of blood comes out of the body.

Due to Pregnancy and Lactation, the woman needs more protein. During pregnancy, the fetus develops in the mother's womb. The development of the fetus depends on the diet of the mother. Therefore, a large amount of protein is needed for the development of the fetus growing in the womb. There is an increased protein demand during pregnancy. In a lactating woman, the baby is nourished by the mother's milk. Therefore, a lactating mother needs more protein than an ordinary woman.

In the state of fever-There is more breakdown of body cells when there is a fever.

Therefore, more protein is needed in the stage of fever or when the fever is cured.

Similarly, in case of burns, accidents, and excessive blood loss, the demand for protein increases.

Age-Physical development is rapid in children. Therefore, they require more protein.

The demand for protein increases during infancy, childhood, and adolescence. The demand for protein is high even in old age.

Protein-Protein is the need of children suffering from energy malnutrition. Similarly, if the protein digestibility in the digestive system is low, the demand for protein also increases.

Importance of protein

1.Muscle development: Protein is essential for muscle maintenance and growth. A deficiency can lead to muscle weakness and wasting.

2.Poor Immune Function: Many immune system components, such as antibodies, are proteins. A lack of protein can weaken the immune system, making the body more susceptible to infections.

3.Edema: Protein helps maintain fluid balance in the body. A deficiency can lead to fluid accumulation in tissues, causing swelling (edema).

4.Hair, Skin, and Nail Issues: Proteins like **collagen are crucial for skin**, hair, and nail health. A deficiency can result in brittle nails, dry skin, and hair problems.

5.Fatigue and Weakness: Protein deficiency can lead to general fatigue and weakness due to reduced muscle function and energy production.

6.Delayed Wound Healing: Proteins are involved in the repair of tissues. A lack of protein can slow down the healing process for wounds and injuries.

7.Reduced Enzyme Activity: Enzymes, which are proteins, are essential for various metabolic reactions. Protein deficiency can disrupt these reactions, affecting overall metabolism.

8.Impaired Growth in Children: Children require adequate protein for growth and development. A deficiency can lead to stunted growth and developmental delays.

9.Hormonal Imbalance: Some hormones, **like insulin and growth hormone, are proteins**. A lack of protein can disrupt hormonal balance, leading to various health issues.

10.Cognitive and Mood Changes: Neurotransmitters, responsible for mood and cognitive function, are made from proteins. Protein deficiency may lead to mood swings, cognitive impairment, and even depression.

Excess of Protein

1.Kidney Strain: Excessive protein intake can put a strain on the kidneys as they are responsible for filtering and eliminating waste products from protein metabolism, including nitrogen. Over time, this increased workload can lead to kidney damage or dysfunction.

2.Dehydration: High-protein diets often result in increased water loss through urine. This can lead to dehydration if adequate fluid intake is not maintained.

3.Mineral Imbalances: High protein intake can disrupt the balance of minerals in the body, particularly calcium. It may lead to increased calcium excretion through urine, potentially contributing to a **risk of osteoporosis and kidney stones**.

4.Digestive Issues: Excess protein can be hard on the digestive system, potentially causing constipation, bloating, and other gastrointestinal discomfort.

5. Increased Risk of Heart Disease: Some high-protein diets, particularly those high in animal protein, may also be high in saturated fats and cholesterol, increasing the risk of heart disease.

6.Weight Gain: Contrary to some beliefs, excessive protein consumption alone may not necessarily lead to weight loss. If the excess protein intake is not balanced with other nutrients and calorie control, it can contribute to weight gain.

7. Bad Breath and Body Odor: Excess protein metabolism can produce ammonia, which can lead to bad breath and unpleasant body odor.

8. Liver Disorders: In rare cases, excessive protein intake can stress the liver and potentially lead to liver disorders.

9.Nutrient Imbalances: Overemphasis on protein intake can sometimes lead to a lack of other essential nutrients, as individuals may prioritize protein-rich foods over a balanced diet containing fruits, vegetables, and whole grains.

CARBOHYDRATES

Carbohydrates are the combination of carbon, oxygen and hydrogen that can be converted into sugars, chemically these are $C_n(H_2O)_n$.

Generally, carbohydrates are divided into three groups according to the sugar group found in them

Monosaccharides (Monosaccharides or Simple Sugars) Mono saccharides are the carbohydrates which are the simplest form of substances that can be digested by the body. They are sweet in flavour and soluble in water. The monosaccharides found in food are called hexoses, which are divided into two groups

Glucose

Fructose

They are converted into simplest sugars by acid, enzymes and heat. These are of three types: (i) sucrose, (ii) maltose, (iii) lactose.

Starch-Plants store carbohydrates in the form of starch, from which they get energy when needed. 65-85% starch is found in cereals, 19.35% in seeds and tubers and 2-10% in fruits and other vegetables. Starch is insoluble in water.

Sources

All types of cereals such as—Wheat, Rice, Millet, Maize, Barley some types of Pulses, Sugar, Jaggery, Honey, Root vegetables, such as potatoes, sweet potatoes, beets, dried fruits; for example, raisins, figs, dates, apples, pears, mangoes, pods, cashews, khoya, cheese etc. are the best means of getting carbs.

Carbohydrate may be:

- . Refined**
- . Unrefined**

Refined means that the food is highly processed. The fiber as well as many of the vitamins and minerals they contain, have been stripped away. Thus, the body processes these carbohydrates quickly and they provide little nutrition although they contain about the same number of calories, refined products are often enriched, meaning vitamins and minerals have been added back to increase their nutritional value. A diet high in simple or

FUNCTIONS OF CARBOHYDRATES

- 1. Providing energy/strength—**The main function of carbs is to give **energy /strength/heat** in the body. Four calories of energy is provided by one gram of carbs. Calories are needed for the **proper functioning of the body, brain and nervous system**, which is obtained from glucose. Therefore, in both these institutions carbs are involved in providing continuous power. Most of the body activities are related to the activities of the brain and nervous system and depend on carbohydrates for energy.
- 2. Making glycogen-**Another function of carbohydrates is to supply the required amount of glycogen in the body. When excess carbohydrate is taken in the body, it converts glycogen into glucose when the amount of glucose is reduced (i) keeps the amount of glucose in the blood normal and (ii) works to give energy/ power in the body. Normally, more than 100 grams of glycogen is stored in the liver of a healthy human. Nerves or muscles store 250 grams of glycogen.

- 3. Protein saving action-**When the body gets less amount of carbs and fat from food items – food items and protein starts working to give power/energy to the body. At this time, protein leaves its function and starts doing the work of carbs and fats, due to which the energy requirement of the body is fulfilled. As soon as the amount of carbs becomes sufficient in the food, the protein becomes free to do its own work. The work of providing strength to the body is considered to be the most important function of the food elements, after which the food elements can do other work.
- 4. Converting to fat-**When the body consumes more amount of carbs than the requirement, all the excess amount gets converted into fat and gets stored in the adipose tissue.
- 5. Functions of Cellulose-**Peristaltic movement comes in the digestive system when cellulose is in sufficient quantity in the diet. Cellulose also helps in water absorption.

6. **Aid in fat metabolism-**Carbohydrates help in complete metabolism of fats and fatty acids in food. Due to the lack of carbohydrates, the metabolism of fats remains incomplete and the amount of acid and ketone bodies in the blood increases, which leads to diseases called acidosis and ketosis.
7. **Functions of Lactose-**Lactose which is found in milk helps in the absorption of calcium in the small intestine. Lactose stays in the stomach for a long time. This leads to the formation of many B vitamins in the stomach. During lactation, the body's glucose is converted into lactose and retains the lactose content of mother's milk.
8. **Maintaining the normal amount of glucose in the blood-**The main function of the carbohydrate substance is to maintain the normal amount of glucose in the blood. The normal amount of glucose in the blood is 80-100 mg glucose per ml of blood. This quantity increases after a meal and when there is a high amount of carbs in the food, a significant increase in the amount of glucose in the blood is seen. It decreases during fasting and disease.

Other functions—The other functions of carbohydrates are:

- (i) To manufacture some amino acids**
- (ii) To provide ribose, a component of nucleic acids**
- (iii) To make galactosides in the brain tissue**
- (iv) To digest food and to make flavouring**
- (v) To provide satisfaction/satiation from food**
- (vi) To produce vitamins by fermenting bacteria.**

EFFECTS OF DEFICIENCY OF CARBOHYDRATE

The amount of carbs in the daily diet should not be reduced. In the absence of carbs and fats, protein starts to provide energy. Due to this, the main function of protein, which is to build muscle, blood, etc., becomes secondary and the work of energy producer becomes main. Due to this the growth and development of the body stops. The following effects are caused by a lack of carbs:

- (1) Loss of weight: Deficiency of carbohydrates leads to the loss of protein and fats. Thus it some times reduce the body mass and weight as well.**
- (2) Feeling tired and weak: Carbohydrate is the primary source of energy for our body. Deficiency of carb make our body tired and weak.**
- (3) Physical development is stunted: Prolong deficiency of carb restrict body development and growth. Marasmus is a diseases that can be seen in a child with car deficiency along with protein and fat.**

(4) Nervousness and irritability in nature: As carbohydrate is the primary energy source, carb deficiency impacts the nervous system and sometimes makes the child more irritable.

(5) Decreased body activity: Body activity depends on the energy obtained from the food. As in the deficiency of carb make body deficient of energy, therefore the activity of the body decreases.

(7) Wrinkling of the skin

(8) Manifestations of malnutrition

(9) Disturbances in the normal functioning of the nervous system

(10) Disorders related to the digestive system.

EFFECTS OF EXCESS OF CARBOHYDRATE

- (1) Excess of carbs in the daily diet leads to obesity. Obesity reduces the flexibility of the body. The working capacity is reduced. Fat people get tired quickly. Obesity leads to diseases like heart disease, diabetes etc.**
- (2) Consumption of more carbs in food leads to the formation of glucose. Excess glucose gets converted into glycogen and stored in the liver. Excess glucose gets converted into fat, which leads to obesity.**
- (3) The food is not digested due to the consumption of more carbs in the diet and due to indigestion vomiting, diarrhea also occur.**
- (4) Eating too much of sugar, jaggery, sweets or chocolate leads to tooth decay.**
- (5) Consumption of excessive carbs leads to the formation of more glucose. But the capacity of storing glucose in the blood is only 80 mg / 100 ml. The remaining glucose either gets converted into glycogen and goes to the liver, muscles, or it starts coming out with urine, which is a symptom of diabetes.**

(6) The pancreas has to work more due to the excess of carbs. A hormone called insulin is released from the the pancreas. This hormone helps in converting glucose into glycogen. The constant release of insulin hormone puts more burden on the pancreas. Therefore, the ability to produce insulin in the desired amount is destroyed, due to which the amount of glucose in the blood remains more than the prescribed amount, which is excreted out of the body through urine. The exit of glucose through urine is called Diabetes or Glucosuria.

DAILY DEMAND OF CARBOHYDRATE

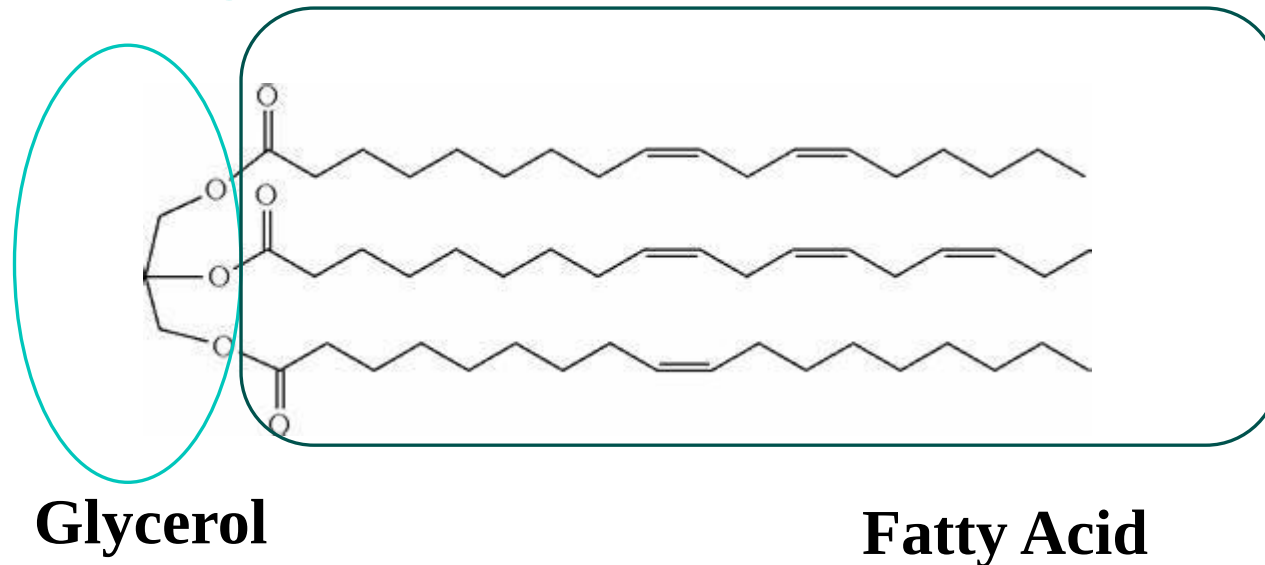
A healthy adult person should take 400–500 grams of carbs in his daily diet. 60-70% of the total body energy demand comes from carbs. Age, gender of the person and the basis of work determines the calorie demand. If the food contains 8 parts of carbs, then at least one part should be protein and one part fat, only then the proper use of protein is done in construction work otherwise protein leaves main work of building the body and starts doing auxiliary work.

People of the rich class get 50-60% of their energy from carbs, while the poor people depend on carbs for energy up to 75-85%. In developed countries America, Japan, Canada like and Russia, people get 40-50% of their energy from carbs and the rest from fat, water (H₂O).

Fats

Fats are oily substances occurring in the animal body and deposited in layers under the skin or around different organs.

Chemically, fats belong to the ester family. Fats are esters formed from the combination of long-chain fatty acids and glycerol.



Fats provide energy of about 9 cal/g. Around 20-30% of total energy is obtained from fats.

Sources of Fats



Table - Fat content in various food items

Sr. No.	Name of the food item (per 100 grams)	fat (in grams)
1.	Ghee and oil	100
2.	Vegetable Ghee	100
3.	Fresh Ghee	99.5
4.	Butter	81.0
5.	Coconut Powder	65.0
6.	Coconut Fresh	42.0
7.	Almonds	59.3
8.	Cashew Nuts	47.0
9.	Sesame	43.3



895	12.
729	13.
600	14.
370	15.
532	16.
530	17.
390	18.
	19.

Name of the food item (per 100 grams)	fat (in grams)	calories (per 100 g)
Almondnut	40.1	360
Ghee from buffalo milk	31.2	271
Cow's Milk	36	4.0
Goat's Milk	50	5.5
Mother's Milk	35	3.9
Fish	28	3.2
Meat	119	13.3
Liver	27	3.0
Wheat Embryo	69	7.4
Millet	45	5.0

FUNCTIONS OF FATS

Fat performs the following functions in the body:

- 1. Providing energy**—A lot of energy is released from fat. Usually 9 calories of energy is released from one gram of fat.
- 2. Storage of energy**—The amount of excess fat in the body is present in the form of adipose tissue below the skin in the form of a layer. At the time of need, this stored fat is used for energy production and when energy is not needed then this fat keeps on accumulating in the fatty fibres, due to which the person becomes fat.
- 3. Regulating body temperature**—A normal layer of fat remains under the skin which acts as a bad conductor of heat in the body. Being an insulator, the temperature of the body remains regulated by the environmental influence.
- 4. Obtaining fat soluble vitamins**—Fat is necessary for the absorption of vitamins A, D, E, K. If the amount of fat in the diet is low, these vitamins will not be able to accumulate, and various deficiency diseases will occur.

5. **Providing essential fatty acids**—Some essential fatty acids are not synthesized by the body, it is necessary to supply them through food. The deficiency of essential fatty acids synthesis leads to various skin diseases. It is necessary to take them in good quantity to keep the body healthy.
6. **To protect the soft parts of the body**-Some essential fatty acids protects soft organs like the kidney, the heart and nervous system from external shocks by forming a deep structure. The highly sensitive organs are also covered with a double layer of fat to protect the soft organs from external shocks.
7. **Prolonged satisfaction**-Fat slows down the secretion of digestive juices. If the digestive juices are secreted late, then the process of digestion will take time. Therefore, food will remain in the digestive organs for a long time, due to which the person feels the satisfaction of food for a longer time and does not feel hungry.
8. **Saving protein**-If energy is not supplied by carbohydrates, then protein also does the work of energy production leaving its main building work. In such a situation, if the amount of fat is present in the body, the fat prevents the protein from being used for energy production, thus saving the protein.

- 9. To make food tasty-**Fat imparts distinctive taste and smell to food elements. Food is prepared and served in the form of various dishes using fat.
- 10. Lubricates the gastric and intestinal tract-**Fat keeps the muscles of the stomach and intestinal tract lubricated (in the form of lubricant) which helps in contraction and release of muscles, conduction of nerve sensations and blood pressure control.

EFFECTS OF DEFICIENCY OF FAT

Due to which the normal growth and development of the body stops:

- 1. Normal Growth Stunted-**Deficiency of fats affect activity of amino acids in the body.
- 2. Skin Becomes Dry Rough and Lustreless-**Oily glands are present in the dermis layer of the skin. These glands secrete oil continuously, due to which the skin looks smooth and glowing. In the absence of fat, these glands do not secrete sufficient amount of sebum (oil), due to which the skin becomes dry, rough and shineless.
- 3. Phrynoderma-**Deficiency of fat causes phrynoderma disease in children and adults. In this disease, sharp pimples appear on the skin of the back, abdomen, thighs and hips.

4. **Decrease in the Capacity of Cells**—The water balance in the body gets disturbed. There is an imbalance in the processes of some enzymes, due to which the speed of cholesterol digestion decreases. Due to this, the amount of cholesterol in the blood increases. Heart diseases occur due to the increase in the amount of cholesterol in the blood.
5. **Essential for Reproduction**-It has been proved by scientific research that Vitamin 'E' is necessary for reproduction. This vitamin is also transported through fat. Due to the lack of fat in the diet, the transport and absorption of vitamin E is not fully done, due to which disturbances in the reproduction process arise.

EFFECTS OF EXCESS OF FAT

The following side effects are seen due to excess of fat:

1. **Increase in Obesity**-Consumption of more fatty food than necessary leads to the formation of adipose fibers in excess, due to which a layer of fat accumulates under the skin in other soft organs like lungs, heart, testes, liver etc and other places of the body. Due to the accumulation of fat layer, obesity increases. Obesity causes many other diseases like diabetes, high blood pressure, heart disease etc.

Diabetes-Excess intake of fats and carbs leads to the formation of excessive amount of glucose. Only a limited amount of glucose can be stored in the blood. The remaining glucose is converted into glycogen and stored in the liver and muscles. Glucose cannot be converted into glycogen due to the low production of insulin. Due to which glucose starts being expelled from the urinary tract along with urine. This is called diabetes.

Heart Related Disease—The amount of cholesterol in the blood increases due to excess of fat. Due to its accumulation, the amount of cholesterol starts depositing on the inner walls of the arteries. Gradually, cholesterol accumulates on the walls of these arteries. This hardens the walls of the arteries and reduces the flow of blood. This increases the blood pressure, its direct effect falls on the heart and this increases the chances of heart attack.

Effects kidney-Due to obesity, a thick layer of fatty fibers accumulates around the kidney and waste material like urea, uric acid, ammonia, toxins etc. are not completely removed and pigment and harmful substances remain in the blood. As a result, the process of filtering blood is not done properly and pigment and harmful substances remain dissolved in the blood itself. Impure blood is toxic to the body, due to which the person dies. Gradually the kidneys stop working

DAILY DEMAND OF FAT

The body should get a good amount of essential fatty acids through food for the smooth completion of all the functions of fat content in the body. For this purpose, 50% of the total aliphatic substances of food should be taken by vegetable oils.

Generally, the requirement of aliphatic substances of every person is fulfilled by getting a certain percentage of calories from fat substances as follows:

Adult (pregnancy)	10-20% calories from fat
Children 1-18 years old	15-20% calories from fat
Infant (one year old)	25-30% calories from fat
Active adults	30-40% calories from fat